Do adults, children, or monkeys use push-down stacks when producing center-embedded sequences?

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Recent findings in artificial grammar tasks have shown that the ability to represent center-embedded sequences is more widespread than initially thought. Adults, young children, non-human primates, and crows can all learn to represent center-embedded sequences and generalize them to novel strings (Ferrigno et al., 2020; Jiang et al., 2018; Liao et al., 2022; Malassis et al., 2020). It is often assumed that humans (and potentially other animals) represent recursive sequences using a push-down stack, a last-in-first-out memory structure similar to computer programs (Fitch & Hauser, 2004; Jiang et al., 2018; Joshi, 1990). However, it is also possible that these sequences could be represented using a queue, a last-in-last-out memory structure.

A push-down stack is a memory architecture that stores items in a *last-in, first-out* order. This type of memory structure is well suited to generate center-embedded structures; the second half of the sequences can be generated by just popping the last item off the stack (e.g., hold A B C in working memory, then retrieve C B A in order from the stack). In contrast, representing a cross-serial sequence with a push-down stack would require additional processing (and an additional stack) to first rearrange the items in the stack to produce a cross-serial sequence (Joshi, 1990). Contrary to these predictions, artificial grammar studies testing the relative difficulty of representing center-embedded and cross-serial sequences often find that there is either no difference, or that center-embedded grammars are harder to learn and generate than cross-serial grammars (De Vries et al., 2012; Öttl et al., 2015; Silva et al., 2022, Uddén et al., 2012). A similar finding has been seen in natural language grammars, such that it was harder for

German speakers to compute the correct meaning of center-embedded sentences than for Dutch speakers to compute the meaning from the semantically identical cross-serial sentences (Bach et al., 1986).

Another potential memory structure that could represent both center-embedded and cross-serial grammars is a queue. A queue is a *first-in*, *first-out* memory structure. A queue memory architecture could easily underlie the production of cross-serial sequences (e.g., hold ABC in working memory queue, start from the beginning of list, and immediately read each item from the beginning to create ABC|ABC. However, producing a *center-embedded sequence* with this type of memory architecture would require additional processing because the order would need to be reversed. Using a queue to generate center-embedded structures also predicts item-by-item response time differences: as you search from the beginning of the queue, the number of items to step through *decreases* as you get closer to the last item needed leading to decreasing item-by-item response times for the second half of a center-embedded structure.

Here, we test 100 US adults, 100 US children, and 4 rhesus macaques on a visual sequence generation task. Participants were trained to click on images one by one in order and then tested their ability to generalize these orders to new combinations of items, completely novel items, and novel lengths (Fig.1). We use the above accuracy and response time signatures from the stack and queue models to test which best fit the response data from the center-embedded and cross-serial sequences.

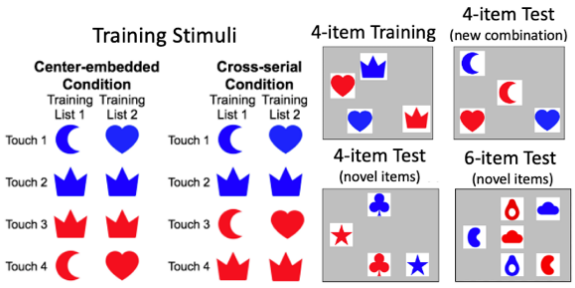

Figure 1. Examples of the two sets of training stimuli for center-embedded and cross-serial orders (depending on condition). After training, participants received the 4-item test trials (new combination), then the 4-item test trials (novel items), and the 6-item test trials all without differential feedback.

Our results show that adults, children, and monkeys can represent and generalize center-embedded and cross-serial structures to novel strings. Furthermore, we find multiple signatures of a queue-like memory structure: faster RTs in cross-serial compared to center-embedded, decreasing RTs for the second half of center-embedded sequences, and higher accuracy for cross-serial sequences. The data and modeling converge on the conclusion that adults,

children, and monkeys use a queue-like memory architecture when generating center-embedded and cross-serial sequences. Although this task differs from natural language in many ways (including meaning, clause structure, and implicit representations, among many others), it may offer one possible explanation for why similar effects have been seen in natural language (Bach et al., 1986). Additionally, this work offers insight into the types of computational mechanisms adults, children, and non-human animals rely on during complex sequencing.

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